Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	*(a)*		Mass defect = $\left(\left(235.01+1.01\right)-\left(91.90+140.89+3(1.01)\right)\right)$ u [or $\left(235.01-\left(91.90+140.89+2\times1.01\right)\right)$] = 0.20 u (1) Energy released = 0.20 $\times$ 931 = 186 MeV [190 MeV] (1) (Accept 3.0 × 10^{-11} J) **UNIT**		2		2	2	
	(b)		**General (G)** G1:General shape of curve [correct asymmetry] & axes labelled G2:Maximum around nucleon number 60 / iron / nickel G3:Stability linked to BE/nuc to stability G4:Nucleons react to move towards maximum (on curve) G5:In doing so there is a loss / reduction in mass G6: Mass loss linked to energy released = $\Delta m\ c^2$ **Fusion (Fu)** Fu1: Smaller nucleon number nuclei combine Fu3: Larger nucleons of larger nucleon number formed: energy released, BE/nuc increases, stability increases, lower mass **Fission (Fi)** Fi1: Larger nucleon number split Fi3: Smaller nucleons of smaller nucleon number formed energy released, BE/nuc increases, stability increases, lower mass Additional point: gradient of curve larger in fusion region than fission resulting in more energy release (per nucleon) **5-6 marks** At least 6 G points (6-8 G) 8 – 11 points *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.*	6			6		

			3-4 marks 5 – 7 points *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.* **1-2 marks** 1 – 4 points *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** *No attempt made or no response worthy of credit.*						
			Question 4 total	**6**	**2**	**0**	**8**	**2**	**0**